

Sub-Project Software Requirements Specification

for

User Identity, Profile and Portfolio Management

Module 1 — National Freelance & Skill Verification Platform

Version 1.0 approved

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1. Introduction

1.1 Purpose

This Sub-Project Software Requirements Specification (SP-SRS) describes the functional and non-functional requirements for the User Identity, Profile and Portfolio Management Module (Module 1) of the National Freelance and Skill Verification Platform. This document is intended to serve as the primary reference for development, testing, and integration activities related to this module. It provides a detailed specification of all software requirements for Module 1 and establishes a common understanding among developers, testers, the integration group, and academic evaluators.

1.2 Document Conventions

This document follows the IEEE 830 Standard for Software Requirements Specifications. The following conventions are used throughout:

- Functional requirements are identified using the prefix FR-PM (e.g., FR-PM-01).
- Non-functional requirements are identified using the prefix NFR-PM (e.g., NFR-PM-01).
- The terms 'shall' denote mandatory requirements, and 'should' denote desirable requirements.
- Higher-level requirements inherit their priority to sub-requirements unless stated otherwise.
- TBD (To Be Determined) is used as a placeholder where information is not yet finalized.

1.3 Intended Audience and Reading Suggestions

This document is intended for the following audiences:

- Developers: Should focus on Sections 3, 4, and 5 to understand functional requirements and system constraints.
- Testers: Should refer to Sections 4 and 5 for test case derivation and validation criteria.
- Integration Group: Should review Section 3 (External Interfaces) and Section 4 (System Features) to understand integration points.
- Project Instructor / Evaluator: May read all sections sequentially for a complete overview of module requirements.
- Project Manager / Team Lead: Should review all sections to ensure completeness and traceability.

It is recommended that all readers begin with Section 2 (Overall Description) for context before proceeding to detailed sections.

1.4 Product Scope

The User Identity, Profile and Portfolio Management Module is a core component of the National Freelance and Skill Verification Platform. This module is responsible for managing user identities and enabling freelancers, clients, and administrators to register, authenticate, and manage their professional presence on the platform.

The module supports multi-role user registration, secure authentication, profile creation and editing, portfolio management, work history tracking, and a review and rating system. It serves as a foundational data provider for other modules including job matching, skill verification, communication, and payment services.

This specification covers Module 1 only and excludes skill assessment, job posting, payment processing, and advanced analytics functionalities, which are addressed in separate module specifications.

1.5 References

- Business Case — National Freelance and Skill Verification Platform (Assignment 1)
- Project Charter — National Freelance and Skill Verification Platform
- Kickoff Meeting Minutes — 26th March 2026
- Sub-Project Management Plan (SPMP) — Module 1, Version 1.0
- Sub-Project Requirements Management Plan (SP-RMP) — Module 1, Version 1.0
- IEEE Std 830-1998 — IEEE Recommended Practice for Software Requirements Specifications
- Module Decomposition Document — National Freelance and Skill Verification Platform (11 Modules)

2. Overall Description

2.1 Product Perspective

The User Identity, Profile and Portfolio Management Module is a sub-system of the larger National Freelance and Skill Verification Platform. It does not function as a standalone product but is tightly integrated with the overall system architecture. The platform aims to create a trusted and transparent digital environment for freelancers and clients in Pakistan's growing gig economy.

Module 1 acts as the identity and profile backbone of the platform. All other modules depend on it for user identity verification, profile data access, and role-based authorization. For instance, the AI Matching Module (Module 4) relies on profile skills data, the Payment Module (Module 7) requires verified user identity, and the Communication Module (Module 6) uses user avatars and role information.

The module will be developed as a web-based system and will interact with a single shared centralized database used across all platform modules, along with authentication services and REST APIs. All modules on the platform share the same PostgreSQL database instance, with each module owning and managing its respective tables. Cross-module data access is coordinated through well-defined REST API endpoints rather than direct cross-module database queries, ensuring loose coupling and maintainability.

2.2 Product Functions

The major functions supported by Module 1 are summarized below:

- User Registration: Multi-role registration for freelancers, clients, and administrators.
- User Authentication: Secure login and logout with session management.
- Profile Management: Create, view, update, and manage detailed professional profiles.
- Portfolio Management: Upload, edit, delete, and display portfolio items and project samples.
- Work History Tracking: Record and display professional experience and completed projects.
- Reviews and Ratings: Allow clients to submit ratings and reviews visible on freelancer profiles.
- Role-Based Access Control: Enforce access restrictions based on user roles.
- API Provision: Expose structured data endpoints for consumption by other modules.

2.3 User Classes and Characteristics

The following user classes are identified for this module:

User Class	Description	Technical Expertise	Primary Functions
Freelancer	Professional offering services on the platform	Low to Medium	Registration, profile & portfolio management, work history
Client / Employer	Individual or organization seeking services	Low to Medium	Registration, profile viewing, submitting reviews
Administrator	Platform staff managing users and content	High	User moderation, profile oversight, access control

2.4 Operating Environment

The module will operate in the following environment:

- Platform: Web-based application accessible via standard web browsers (Chrome, Firefox, Edge, Safari).
- Client Side: Any device with a modern web browser and internet connectivity (desktop, tablet, mobile).
- Server Side: Node.js / React-based backend with RESTful API architecture.
- Database: Shared centralized PostgreSQL database instance used across all platform modules. Module 1 owns and manages the Users, Profiles, Portfolio Items, Work History, and Reviews tables within the shared schema.
- Hosting: Cloud-hosted or institutional server environment.
- Authentication: JWT (JSON Web Token) based secure authentication.

2.5 Design and Implementation Constraints

- The module must be developed within a single academic semester (March–May 2026).
- The team consists of four members with limited development time.
- Only open-source and freely available tools and frameworks may be used.
- The module must conform to the overall system architecture defined by the Integration Group.
- OTP-based verification is out of scope for this module.
- All APIs must follow RESTful conventions and return JSON responses for interoperability.
- The system must not implement real-world financial or legal verification services.

2.6 User Documentation

- User registration and profile management guide (to be prepared during development phase).
- API documentation for integration with other modules.
- Inline help text and form validation messages within the user interface.

2.7 Assumptions and Dependencies

Assumptions:

- Users will provide accurate and complete registration information.
- All platform users will have access to internet-enabled devices with modern browsers.
- Other modules will be developed concurrently and will be available for integration testing.
- Standard web technologies (React, Node.js, REST APIs) are sufficient for implementation.

Dependencies:

- Module 2 (Skill Testing & Certification): Writes verified skill badges to the shared database; Module 1 reads and displays them on user profiles.
- Module 8 (Dispute & Admin): Writes moderation flags, suspensions, and ban decisions to the shared database; Module 1 reads and enforces them on user accounts.
- Module 10 (Social Impact): Writes social impact badges and impact points to the shared database; Module 1 reads and displays them on user profiles.
- Module 11 (Gamification): Writes trust scores and reputation badges to the shared database; Module 1 reads and displays them on user profiles.

3. External Interface Requirements

3.1 User Interfaces

The module shall provide a web-based graphical user interface (GUI) with the following characteristics:

- Registration and Login pages with form validation and error messaging.
- A Profile Dashboard allowing users to view and edit their profile information.
- A Portfolio Section with upload, preview, and management capabilities.
- A Work History section for recording and displaying past projects.
- A Reviews and Ratings display panel visible on freelancer profiles.
- Role-specific navigation and access controls ensuring freelancers, clients, and admins see appropriate views.
- Responsive design supporting desktop and mobile browsers.
- Consistent UI conventions aligned with the overall platform design language.

3.2 Hardware Interfaces

The module does not directly interface with specific hardware components. It operates as a web application accessed through standard internet-connected devices. The following hardware assumptions apply:

- Client devices must support a modern web browser with JavaScript enabled.
- Server hardware must support the hosting of a Node.js/React application with database connectivity.
- File upload functionality requires sufficient server storage for portfolio attachments.

3.3 Software Interfaces

The module will interface with the following software components and services:

Software Component	Version	Purpose	Interface Type
React.js	18+	Frontend UI rendering	Internal
Node.js / Express	18+	Backend API server	Internal
PostgreSQL / MySQL	Latest	User and profile data storage	Internal
JWT Library	Latest	Authentication tokens	Internal
Module 2 (Skill Testing)	—	Write verified skill badges to shared DB; Module 1 reads them	Shared Database
Module 8 (Admin & Dispute)	—	Write moderation flags/suspensions to shared DB; Module 1 reads them	Shared Database
Module 11 (Gamification)	—	Write trust scores and badges to shared DB; Module 1 reads them	Shared Database

3.4 Communications Interfaces

- All client-server communication shall use HTTPS protocol to ensure data security.
- RESTful APIs shall communicate using JSON formatted request and response bodies.
- Cross-Origin Resource Sharing (CORS) policies shall be configured to permit access from authorized platform domains only.
- HTTP status codes shall be used consistently (200, 201, 400, 401, 403, 404, 500) to communicate API outcomes.
- All platform modules, including Module 1, shall share a single centralized PostgreSQL database instance. Each module is responsible for its own tables within the shared schema. Cross-module data sharing shall be performed via REST API calls rather than direct cross-schema database queries, ensuring loose coupling between modules.

4. System Features

4.1 User Registration and Authentication

4.1.1 Description and Priority

This feature enables users to create accounts and securely log in to the platform. It is the entry point for all platform interactions and is of High priority as the entire system depends on it.

4.1.2 Stimulus/Response Sequences

- User navigates to the registration page → System displays a registration form with role selection (Freelancer / Client / Admin).
- User fills in required details and submits → System validates inputs, creates account, and displays confirmation.
- User navigates to login → System displays login form → User enters credentials → System authenticates and issues a JWT token → User is redirected to their dashboard.
- User submits invalid credentials → System displays an appropriate error message.
- User logs out → System invalidates the session token and redirects to the login page.

4.1.3 Functional Requirements

- **FR-PM-01:** The system shall allow users to register using a unique email address and password.
- **FR-PM-02:** The system shall support three registration roles: Freelancer, Client, and Administrator.
- **FR-PM-03:** The system shall validate all registration inputs including email format, password strength, and required fields.
- **FR-PM-04:** The system shall securely authenticate registered users using email and password credentials.
- **FR-PM-05:** The system shall issue a JWT token upon successful login for session management.
- **FR-PM-06:** The system shall allow users to log out, invalidating their session token.
- **FR-PM-07:** The system shall enforce role-based access control, granting each user type access only to permitted features.

4.2 Profile Management

4.2.1 Description and Priority

This feature enables users to create and manage detailed professional profiles. For freelancers, this is the primary means of presenting their professional identity to clients. Priority: High.

4.2.2 Stimulus/Response Sequences

- Logged-in user navigates to 'Edit Profile' → System displays a profile form pre-filled with existing data → User updates fields and saves → System validates and stores updated profile.
- Visitor or client navigates to a freelancer's public profile → System retrieves and displays the profile.

- User attempts to edit another user's profile → System denies access and returns an authorization error.

4.2.3 Functional Requirements

- **FR-PM-08:** The system shall allow users to create a profile including name, bio, location, profile picture, and skill tags.
- **FR-PM-09:** The system shall allow users to edit and update their profile information at any time.
- **FR-PM-10:** The system shall display freelancer profiles to clients and visitors in a structured, readable format.
- **FR-PM-11:** The system shall support uploading a profile picture in common image formats (JPG, PNG).
- **FR-PM-12:** The system shall prevent users from editing profiles belonging to other users.
- **FR-PM-13:** The system shall display received skill badges and trust indicators received from other modules on the profile.

4.3 Portfolio Management

4.3.1 Description and Priority

This feature allows freelancers to upload and showcase their past work, project samples, and external links. It is a key trust-building mechanism for the platform. Priority: High.

4.3.2 Stimulus/Response Sequences

- Freelancer navigates to the Portfolio section → Clicks 'Add Item' → Fills in title, description, file/link → Submits → System stores and displays the portfolio item.
- Freelancer selects an existing item → Clicks 'Edit' → Updates details → Saves → System updates the record.
- Freelancer clicks 'Delete' on a portfolio item → System prompts confirmation → On confirm, removes the item.
- Client views a freelancer's profile → Portfolio section is displayed with all uploaded items.

4.3.3 Functional Requirements

- **FR-PM-14:** The system shall allow freelancers to upload portfolio items including files (PDF, images) and external URLs.
- **FR-PM-15:** The system shall allow freelancers to add a title and description to each portfolio item.
- **FR-PM-16:** The system shall allow freelancers to edit or delete their portfolio items.
- **FR-PM-17:** The system shall display all portfolio items on the freelancer's public profile.
- **FR-PM-18:** The system shall enforce that only the portfolio owner may add, edit, or delete their items.

4.4 Work History Tracking

4.4.1 Description and Priority

This feature provides a structured record of a freelancer's professional experience and completed projects. Priority: Medium (Should Have).

4.4.2 Stimulus/Response Sequences

- Freelancer navigates to 'Work History' → Clicks 'Add Entry' → Fills in project name, client, duration, description → Saves → System records the entry.
- Freelancer selects an entry and edits or deletes it → System updates accordingly.
- Client views a freelancer's profile → Work history section displays all recorded entries.

4.4.3 Functional Requirements

- **FR-PM-19:** The system shall allow freelancers to add work history entries including project name, client name, duration, and description.
- **FR-PM-20:** The system shall allow freelancers to edit or delete their work history entries.
- **FR-PM-21:** The system shall display work history entries on the freelancer's public profile in reverse chronological order.

4.5 Reviews and Ratings System

4.5.1 Description and Priority

This feature allows clients to submit ratings and written reviews for freelancers after project completion. Reviews are displayed on the freelancer's public profile and contribute to their credibility. Priority: Medium (Should Have).

4.5.2 Stimulus/Response Sequences

- Client navigates to a freelancer's profile → Clicks 'Leave a Review' → Submits a star rating (1–5) and written feedback → System saves and displays the review.
- System calculates and updates the average rating displayed on the profile.
- Freelancer views their profile → Sees all reviews and average star rating.

4.5.3 Functional Requirements

- **FR-PM-22:** The system shall allow clients to submit a star rating (1 to 5) for a freelancer.
- **FR-PM-23:** The system shall allow clients to submit a written review alongside the rating.
- **FR-PM-24:** The system shall calculate and display the average rating on the freelancer's profile.
- **FR-PM-25:** The system shall display all submitted reviews on the freelancer's public profile.
- **FR-PM-26:** The system shall restrict review submission to registered clients only.

5. Other Non-Functional Requirements

5.1 Performance Requirements

- NFR-PM-01: Profile pages shall load within 3 seconds under normal network conditions.
- NFR-PM-02: API responses for profile and portfolio data retrieval shall not exceed 2 seconds.
- NFR-PM-03: The system shall support at least 100 concurrent users without degradation in performance during the academic demonstration phase.

5.2 Safety Requirements

- NFR-PM-05: The system shall not expose sensitive user data (passwords, tokens) in API responses or client-side storage.
- NFR-PM-06: The system shall implement input sanitization to prevent injection attacks on all form inputs.
- NFR-PM-07: All destructive operations (profile deletion, portfolio removal) shall require explicit user confirmation.

5.3 Security Requirements

- NFR-PM-08: All passwords shall be stored as hashed values using a secure hashing algorithm (e.g., bcrypt).
- NFR-PM-09: Authentication tokens (JWT) shall have a defined expiry duration and shall be invalidated on logout.
- NFR-PM-10: All communications between client and server shall be transmitted over HTTPS.
- NFR-PM-11: Role-based access control shall be enforced on all API endpoints to prevent unauthorized data access.
- NFR-PM-12: File uploads shall be validated for file type and size to prevent malicious uploads.

5.4 Software Quality Attributes

- Usability: The interface shall be intuitive and navigable without technical training. Key tasks (registration, profile update) shall be completable within 5 user actions.
- Reliability: The module shall maintain 99% uptime during the project demonstration period.
- Maintainability: Code shall follow modular design principles with clear documentation to support future updates.
- Portability: The module shall function correctly on the latest versions of Chrome, Firefox, Edge, and Safari.
- Scalability: The system architecture shall support future expansion of features without requiring full redesign.
- Interoperability: All APIs shall follow RESTful standards and return JSON-formatted responses to ensure compatibility with other platform modules.

5.5 Business Rules

- Only registered and authenticated users may access profile management features.

- Freelancers may not submit reviews for other freelancers; only clients may submit reviews.
- Administrators have the authority to view and moderate all user profiles.
- A user may only have one account per email address on the platform.
- Profile visibility is public by default; all platform users and visitors may view freelancer profiles.

6. Other Requirements

The following additional requirements apply to Module 1:

- **Database Requirements:** All user data, profile information, portfolio items, work history entries, and reviews shall be persisted in a relational database. Appropriate indexing shall be applied to support efficient querying.
- **Localization:** The module shall be developed in English as the primary language. Internationalization support is not required for the current version.
- **Data Retention:** User data shall be retained for the lifetime of the platform. Deleted accounts shall have their personal data anonymized rather than permanently removed.
- **Integration Requirements:** All platform modules share a single centralized PostgreSQL database. Module 1 owns and manages its tables (Users, Profiles, Portfolio Items, Work History, Reviews) within this shared schema. Module 1 shall expose well-documented REST API endpoints to enable other modules to access its data without direct database coupling, including but not limited to Modules 2, 3, 4, 6, 7, 8, 9, and 11.
- **Legal:** No real-world identity verification (e.g., CNIC, passport) is required in this academic version. Compliance with real-world data protection regulations is out of scope.

Appendix A: Glossary

Term / Acronym	Definition
SP-SRS	Sub-Project Software Requirements Specification
SPMP	Sub-Project Management Plan
SP-RMP	Sub-Project Requirements Management Plan
FR-PM	Functional Requirement — Profile Management (Module 1)
NFR-PM	Non-Functional Requirement — Profile Management (Module 1)
JWT	JSON Web Token — a compact, URL-safe token used for authentication
REST API	Representational State Transfer Application Programming Interface
RBAC	Role-Based Access Control
WBS	Work Breakdown Structure
TBD	To Be Determined
HTTPS	Hypertext Transfer Protocol Secure
JSON	JavaScript Object Notation — a lightweight data interchange format
UI	User Interface
OTP	One-Time Password (explicitly out of scope for this module)

Appendix B: Analysis Models

The following analysis models support the requirements defined in this document.

B.1 Use Case Summary

Use Case ID	Use Case Name	Primary Actor	Related Requirements
UC-01	User Registration	Freelancer / Client / Admin	FR-PM-01 to FR-PM-03
UC-02	User Login	All Users	FR-PM-04 to FR-PM-06
UC-03	Manage Profile	Freelancer / Client	FR-PM-08 to FR-PM-13
UC-04	Manage Portfolio	Freelancer	FR-PM-14 to FR-PM-18
UC-05	Track Work History	Freelancer	FR-PM-19 to FR-PM-21
UC-06	Submit Review	Client	FR-PM-22 to FR-PM-26
UC-07	View Profile	Client / Visitor	FR-PM-10, FR-PM-13

B.2 Requirements Traceability Matrix

Req ID	Description	Priority	Feature	Status
FR-PM-01	User registration via email	Must Have	4.1	Pending
FR-PM-02	Multi-role registration support	Must Have	4.1	Pending
FR-PM-03	Input validation on registration	Must Have	4.1	Pending
FR-PM-04	Secure user authentication	Must Have	4.1	Pending
FR-PM-05	JWT token issuance on login	Must Have	4.1	Pending
FR-PM-06	User logout functionality	Must Have	4.1	Pending
FR-PM-07	Role-based access control	Must Have	4.1	Pending
FR-PM-08	Profile creation	Must Have	4.2	Pending
FR-PM-09	Profile editing	Must Have	4.2	Pending
FR-PM-10	Public profile display	Must Have	4.2	Pending
FR-PM-11	Profile picture upload	Must Have	4.2	Pending
FR-PM-12	Profile edit authorization	Must Have	4.2	Pending
FR-PM-13	Badges display on profile	Should Have	4.2	Pending
FR-PM-14	Portfolio item upload	Must Have	4.3	Pending
FR-PM-15	Portfolio item title & desc	Must Have	4.3	Pending
FR-PM-16	Portfolio edit/delete	Must Have	4.3	Pending
FR-PM-17	Portfolio public display	Must Have	4.3	Pending

Req ID	Description	Priority	Feature	Status
FR-PM-18	Portfolio owner authorization	Must Have	4.3	Pending
FR-PM-19	Add work history entry	Should Have	4.4	Pending
FR-PM-20	Edit/delete work history	Should Have	4.4	Pending
FR-PM-21	Work history display	Should Have	4.4	Pending
FR-PM-22	Star rating submission	Should Have	4.5	Pending
FR-PM-23	Written review submission	Should Have	4.5	Pending
FR-PM-24	Average rating calculation	Should Have	4.5	Pending
FR-PM-25	Reviews display on profile	Should Have	4.5	Pending
FR-PM-26	Restrict reviews to clients	Should Have	4.5	Pending

Appendix C: To Be Determined List

The following items remain to be determined and will be resolved during the design and development phases:

TBD ID	Description	Expected Resolution Date
TBD-01	Final technology stack confirmation (frontend framework, database engine)	2 April 2026
TBD-02	Maximum file size limit for portfolio uploads	2 April 2026
TBD-03	Exact API endpoint specifications agreed with Integration Group	5 April 2026
TBD-04	Profile picture storage mechanism (local vs cloud storage)	2 April 2026